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Introduction

The flow control system supplies sufficient fresh air to the Air Conditioning Units (ACUs) and to the trim air lines. The flow control system also controls the air in the foot ventilation zone of the pilot and copilot.

General description

The flow control system is a part of the Environmental Control System (ECS). It measures and controls the quantity of air that flows to the ACUs.

The primary functions of the flow control system are:

- To convert the ozone to oxygen
- To supply hot air to the trim air system lines
- To supply sufficient fresh air to the ACUs
- To isolate the ACUs from the pneumatic system
- To control the supply of air for ventilation in the pilot and copilot foot zone.

The flow control system includes the components that follow:

- The Flow Sensor Venturi (FSV)
- The ozone converters
- The Pack Inlet Flow Sensors (PIFSs)
- The Pack Inlet Pressure Sensors (PIPSs)
- The Footwell Vent Shutoff Valves (FVSOVs)
- The Flow Control Valves (FCVs).

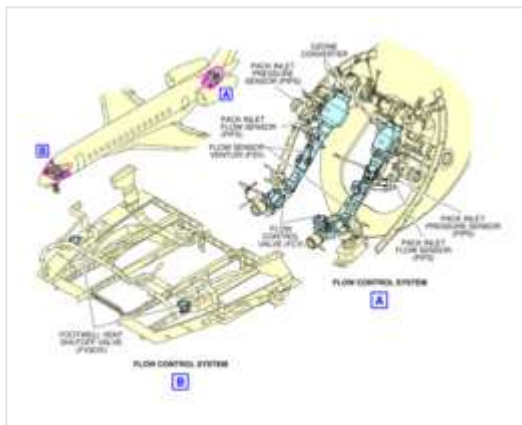


Fig 1 Flow control system - Component location

The flow control system receives the bleed air from the bleed air system. The ozone converters convert the ozone in the bleed air to oxygen by a catalysis effect. This air is sent through the FSVs where the PIPs and the PIFs measure the ACUs inlet pressure and differential pressure. From the FSVs, the air goes to trim air system lines and to the ACUs through the FCVs. For more information about the trim air lines, refer to ATA 21-61-00 Temperature control system.

The pressure and differential pressure values from the PIPs and PIFs are sent to the Integrated Air System Controllers (IASCs) for airflow calculation. The IASCs thus control the quantity of airflow with the FCVs. The FVSOVs control the airflow for ventilation in the flight compartment footwell zone.

The flow control system is automatically controlled and monitored by the IASCs. The flight crew inputs are through the PACK FLOW and L/R PACK switches on the BLEED/AIR COND control panel. The position of the FCVs and airflow lines status are shown on the BLEED synoptic page. The Crew Alerting System (CAS) messages are shown on the EICAS page. The fault messages and real time parameters related to the system are available on the Onboard Maintenance System (OMS) pages.

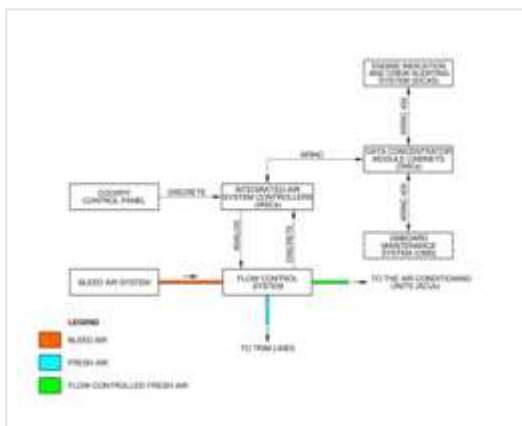


Fig 2 Flow control system - Block diagram

The flow control system have direct interfaces with the system that follow:

- The aircraft electrical system (24-00-00)
- The bleed air system (ATA 36-11-00)
- The air conditioning system (ATA 21-52-00)
- The integrated air system controller (21-91-00)
- The temperature control system (ATA 21-60-00)
- The bleed-air leak-detection system (ATA 36-22-00).

FSV

Two FSVs are installed between the ozone converters and the FCVs in the aft equipment bay at FS1031.00. The FSVs have a venturi throat that creates a pressure difference between the upstream and downstream flow. This pressure difference is sensed by the PIFSSs through sense lines and sent to the IASCs for ACUs inlet airflow calculation.

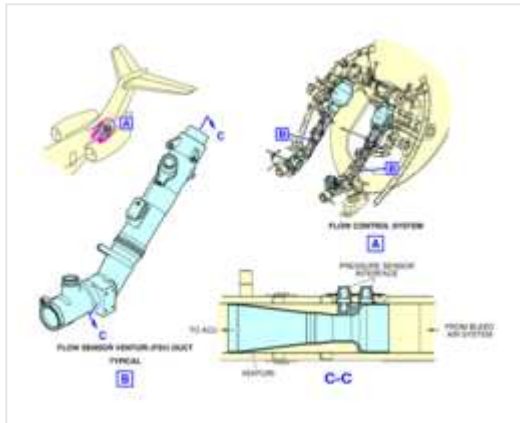


Fig 3 FSV - Component location

Ozone converters

Two ozone converters (one in each bleed line of the ECS) are installed downstream of the precooler ECS and upstream of the FSVs in the aft equipment bay at FS1071.00. The ozone converters convert the ozone molecules contained in the bleed air into oxygen molecules by a catalysis effect.

The ozone converter includes:

- A body with 3 in. (76,20 mm) flanges at the inlet and outlet
- A metallic ceramic core installed in the body which includes an aluminum alloy matrix with platinum and palladium
- A double mounting bracket.

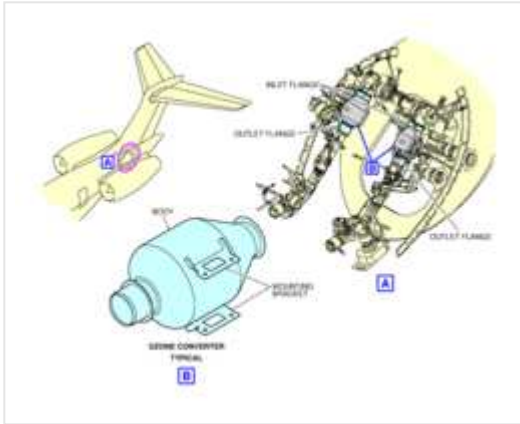


Fig 4 Ozone converter - Component location

PIFS

Two PIFSs are connected to the FSVs with sense lines. The left PIFS is powered by the DC ESS 1 and the right PIFS is powered by the ESS 2 bus. The sensors are made of a sensing diaphragm which is sensitive to differential pressure.

Each PIFS includes:

- Two pressure input ports
- An electrical circuit
- An interval voltage regulator
- A DC amplifier.

The electrical circuit converts the differential pressure into an electrical value. The interval voltage regulator and a DC amplifier give an output voltage in proportion to the differential pressure. The data is sent to the IASCs for ACU inlet flow calculation.

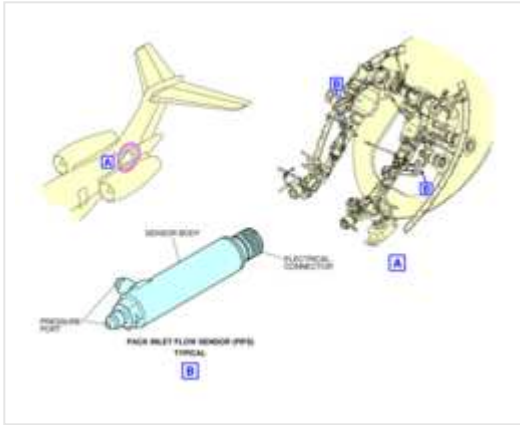


Fig 5 PIFS - Component location

PIFS

Two PIFSs are connected to the FSVs with sense lines. The left PIFS is powered by the DC ESS 1 and the right PIFS is powered by the DC ESS 2. The PIFSs are made of a sensing diaphragm which is sensitive to the absolute pressure.

Each PIFS includes:

- A pressure input port
- An electrical circuit
- An internal voltage regulator
- A DC amplifier.

The electrical circuit converts the pressure into an electrical value. The internal voltage regulator and a DC amplifier give an output voltage in proportion to the absolute pressure. The data is sent to the IASCs for ACUs inlet flow calculation.

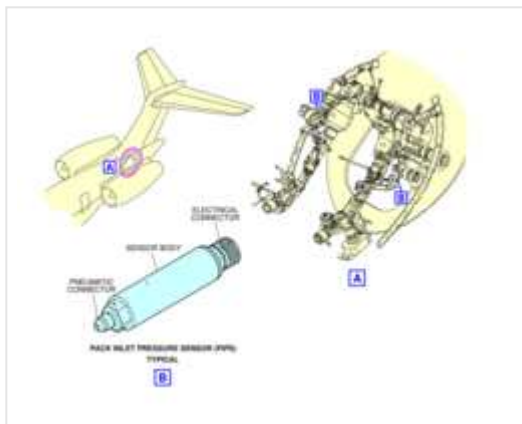


Fig 6 PIPS - Component location

FVSOV

The FVSOVs are installed in the cockpit distribution lines, connected to the low pressure ducting in the branch that supplies the air to the pilot/copilot feet area at FS205L and FS205R. The FVSOVs are used to stop the flow supplied to the pilot and copilot feet area for comfort.

The left FVSOV is powered by the DC ESS bus 1 (SPDA 1) and the right FVSOV is powered by the DC ESS bus 2 (SPDA 2). The FVSOV is a 1.50 in. (38,10 mm) diameter butterfly valve with aluminum body which is operated by an electrical actuator and have a Fully Closed (FC) and Fully Open (FO) micro switches that send discrete signals to the related DMC. The FVSOVs are controlled by the foot vent control switch in the flight compartment.

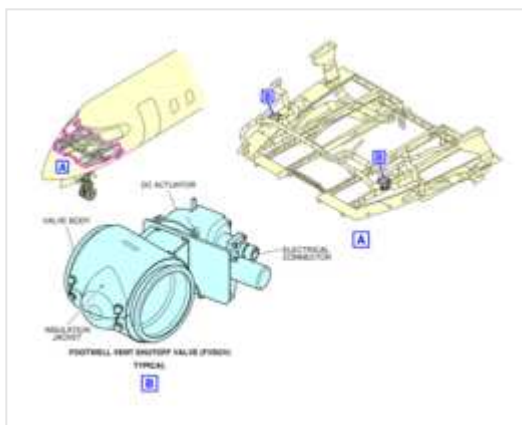


Fig 7 FVSOV - Component location**FCV**

The FCVs are used by the IASCs to control the airflow and pressure to the ACUs and to isolate the ACUs from the pneumatic system.

Each FCV includes:

- A butterfly valve which has a body with a valve plate on an inclined axis, operated by a pneumatic actuator. This body has an upstream pressure inlet to supply the air to the regulator.
- A pneumatic actuator which has a piston or diaphragm and a return spring that closes the valve.
- A Fully Closed (NFC) switch used by the IASCs for position monitoring.
- A regulator which has a valve downstream-pressure limiter,
- A torque motor that modulates the valve position between the fully open and fully closed positions.
- A pressure reducer that gives a constant pressure to the pneumatic actuator.

The valve torque motor is controlled by the IASC. The valve can be closed as follows:

- During the engine start sequence, to make sure that the maximum flow is available to the engine starters,
- If the PACK OFF is selected by the flight crew on the BLEED control panel,
- During ditching, if the aircraft altitude is less than 15000 ft (4572 m),
- If overheat is detected in the bleed duct or in ACUs discharge or compressor discharge conditions,
- If low pressure is detected in ACUs discharge conditions,
- If no power is available to the torque motor or if there is no upstream pressure, or
- If the AUX PRESS is set to ON flight crew on the BLEED control panel.

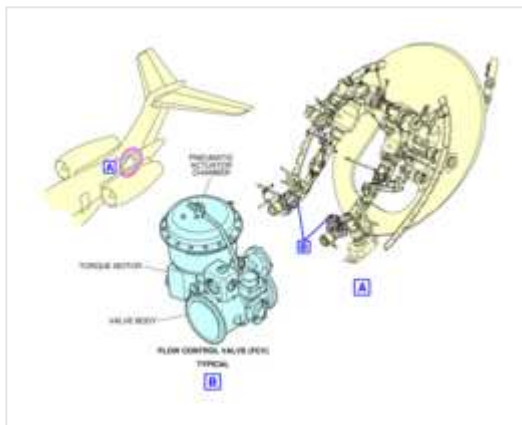


Fig 8 FCV - Component location

System operation

The flow control system receives bleed air from the bleed air system at regulated pressure and temperature. The bleed air is supplied by the engines or the APU or the High Pressure Ground Connection (HPGC) on ground.

The bleed air flows through the ozone converters where the ozone molecules are converted to oxygen molecules by a catalysis effect. The oxygen converted bleed air then flows through the venturi of the FSVs where the PIPs and PIFSs measure the differential pressure and send the data to the IASCs for mass airflow calculation. The IASCs control the position of the FCVs in relation to the calculated mass airflow to supply the sufficient quantity of fresh air to the ACUs. The air from the FSVs is also supplied to the trim air system. The FVSOVs in the cockpit footwell zone receive air from the cockpit low-pressure ducting. These FVSOVs have piccolo ducts for ventilation of the pilot and copilot feet area in the cockpit.

The flow demand is related to:

- The cabin altitude
- The pilot selection (high flow)
- The number of ACUs in operation
- The zone temperature error (difference between zone selection and actual zone temperature).

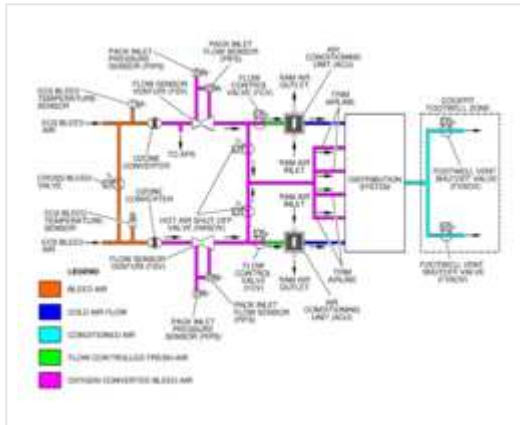


Fig 9 Flow control system - System operation

Control and display

The flow control system is controlled and monitored by the IASCs. The flight crew controls are through the L/R PACK switches and PACK FLOW switch on the BLEED/AIR COND control panel.

- L PACK: During normal operation, the switch is blank.
 - OFF: When pressed in, the left FCV closes to turn off the left ACU and the OFF label on the switch illuminates.
 - FAIL: When the left ACU fails, the FAIL label on the switch illuminates amber and the left FCV closes to turn off the left ACU.
- R PACK: During normal operation, the switch is blank.
 - OFF: When pressed in, the right FCV closes to turn off the right ACU and the OFF label on the switch illuminates.
 - FAIL: When the right ACU fails, the FAIL label on the switch illuminates amber and the right FCV closes to turn off the right ACU.
- PACK FLOW: During automatic flow control mode, the switch is blank.
 - HI: When pressed in, the flow control system switches high flow mode and the HI label on the switch illuminates white.

The control for the flight crew foot ventilation zone is on the pilot/copilot foot vent control panel.

- FOOT VENT switch:
 - OFF: The foot vent is fully closed.
 - ON: The foot vent is fully open.

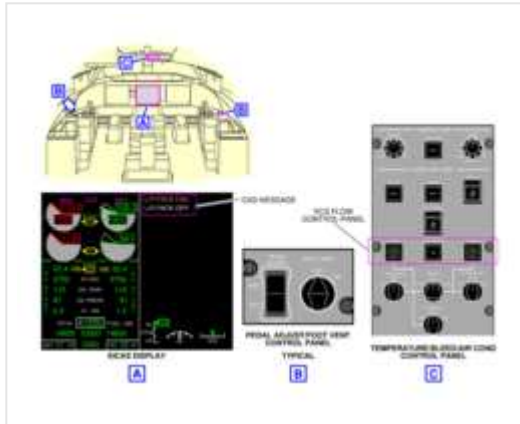


Fig 10 Flow control system - Control and display

The status or any malfunction of the FCVs is shown on the BLEED synoptic page and CAS messages are shown on the EICAS page. The fault messages are shown on the View Fault Messages page of the OMS while the flow control system real time parameters are shown on the Perform LRU System Operation page.

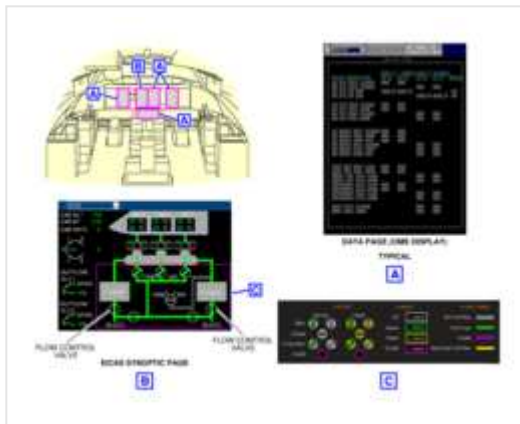


Fig 11 Flow control system - Control and display

System interface

The flow control system interfaces are as follows:

- IASCs: The PIFS sends differential pressure analog signals and the PIPS sends absolute pressure analog signals to for ACU inlet airflow calculation. The FCV torque motor is controlled by the IASCs through an electrical analog signal interface. When the FCV is closed, the IASC receives a ground discrete signal from the FCV fully closed micro switch. From the flight compartment, the L/R PACK switches have a ground/open discrete interface with the IASC 1 and 2 respectively.
- Avionics: The DMCs receive the FOOT VENT switch selection, FVSOVs fully open/closed signals, and the PACK FLOW switch selection through ground/open discrete signals. The IASCs data related to the flow control system is received on an ARINC 429 bus. This data is processed for CAS, status and fault messages annunciations on the EICAS page and OMS page.
- Electrical system: The left PIFS and PIPS are supplied with 28V DC by the DC ESS BUS 1 and the right PIFS and PIPS are supplied by the DC ESS BUS 2 from DC EPC 2.



Fig 12 Flow control system - System interface

[Click here to view the Aircraft Schematic Diagram Publication \(ASDP\).](#)

System monitoring

During normal operation, the IASCs monitors the flow control system output through Continuous Built-In Tests (CBITs) for failures such as out of range, drifts and failed open/failed closed conditions. The IASCs also compare sensors outputs with the thresholds for overpressure/underpressure conditions monitoring.

The flow control system indications are given as EICAS messages as follows:

Table 1 - CAS and INFO messages

Message	Type	Description
L/R PACK FAIL	Caution	- Failure of the left or right FCV in the closed position (when commanded open). - Failure of the left or right FCV in the open position (when commanded closed).
L-R PACK FAIL	Caution	Failure of the left pack and right packs.
L/R PACK OFF	Status	The left/right pack is set to OFF and confirmed commanded OFF by the IASCs.
PACK FLOW HI	Status	The ACU is set to the HI flow mode.
21 AIR COND / PRESS - L PACK INLET FLOW SENSOR INOP	Info	Failure of the left PIFS.
21 AIR COND / PRESS - R PACK INLET FLOW SENSOR INOP	Info	Failure of the right PIFS.
21 AIR COND / PRESS - L PACK INLET PRESS SENSOR INOP	Info	Failure of the left PIPS.
21 AIR COND / PRESS - R PACK INLET PRESS SENSOR INOP	Info	Failure of the right PIPS.
21 L PACK FAIL - FLOW CTRL VALVE INOP	Info	Failure of the left FCV.
21 R PACK FAIL - FLOW CTRL VALVE INOP	Info	Failure of the right FCV.

Maintenance information points

- In some operating conditions, the HI mode is not available because of powerplant limits.
- The ACUs are reset if the L PACK or R PACK are pushed to OFF then to ON again.

Component location

Table 2 - Component location index

Ident	Description	Location/Zone	Maintenance	Parts	Wiring
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Ident	Description	Location/Zone	Maintenance	Parts	Wiring
HP55	Right FVSOV	FS205R/Zone 221/222	AMP21-51-11	IPD21-51-11	AWM21-51-00
HP56	Left FVSOV	FS205L/Zone 221/222	AMP21-51-11	IPD21-51-11	AWM21-51-00
MT127	Left PIFS	FS1071L/Zone 331/332	AMP21-51-07	IPD21-51-07	AWM21-51-00
MT128	Right PIFS	FS1071R/Zone 331/332	AMP21-51-07	IPD21-51-07	AWM21-51-00
MT131	Left PIPS	FS1054L/Zone 331/332	AMP21-51-09	IPD21-51-09	AWM21-51-00
MT132	Right PIPS	FS1054R/Zone 331/332	AMP21-51-09	IPD21-51-09	AWM21-51-00
V18	Left FCV	FS983L/Zone 331/332	AMP21-51-13	IPD21-51-13	AWM21-51-00
V19	Right FCV	FS983R/Zone 331/332	AMP21-51-13	IPD 21-51-13	AWM21-51-00
-	Ozone converters	FS1071.00/Zone 331/332	AMP21-51-05	IPD 21-51-05	-
-	FSV	FS1031.00/Zone 331/332	AMP21-51-01	IPD21-51-01	-